

Autonomy Workload — Data Annex

The computational pipelines, unit-cost derivations and latency model behind every published figure

The companion documents state an argument. This one states the inputs. It enumerates the nineteen algorithm stages that constitute the modeled autonomy pipeline, derives the arithmetic and memory cost of each from algorithm structure at a stated configuration, specifies the sensor suite and update rates of all eighteen mission profiles, and gives the service-time and occupancy model that converts those costs into latency. Nothing is asserted here that is not itemized in the accompanying workbook, and nothing in the companion documents is derived from anything other than this model.

The word that matters is forward. Every unit cost is built up from the structure of the algorithm — window sizes, disparity counts, FFT lengths, parameter counts, iteration counts — at a configuration stated in the open. No unit cost is fitted to a published total. This distinction is the whole point of the document: it means the agreement between this model and the previously published regime figures, reported in Section 6, is a check rather than a construction.

A reviewer wishing to attack these numbers should go to the Unit cost derivation sheet of the workbook first. Every line there is a named term with its arithmetic, and a disagreement about any one of them propagates mechanically to the summary. Section 7 is the list of places where we believe the model is most exposed, including the seven stages where this derivation and the earlier published model disagree by more than a factor of two.

1 · THE MODELED PIPELINE

The pipeline is seven tiers containing nineteen costed stages, of which any given mission profile instantiates between four and nineteen. A profile that omits a stage omits it explicitly — a warehouse AMR runs no radar, an interceptor runs no vision-language model — and the omission is visible in the mission configuration sheet rather than buried in a coefficient.

Tier	Stage	Costed per	Precision class	Ops per call	Op per byte
T1	LiDAR deskew + voxel downsample	per input point	C 100%	170	0.8
T1	Stereo rectify + SGM disparity	per stereo frame-pair	A 85% C 15%	6.4 G	3.9
T1	Radar FMCW post-processing	per radar scan	A 70% C 30%	939,524,096	3.1
T1	Mono camera front-end	per pixel	C 100%	40	10.0
T2	LiDAR-inertial odometry	per input point	C 100%	6,144	5.9
T2	Visual front-end (per camera)	per pixel	A 60% C 40%	124	7.0
T2	Windowed bundle adjustment	per solve	C 100%	168,815,000	44.5
T3	TSDF integration	per measured point	C 100%	3,708	3.1
T3	ESDF propagation	per map update	C 100%	780,000,000	49.2
T4	Open-vocabulary detection	per inference	A 85% B 15%	90.3 G	529.7
T4	Onboard vision-language reasoning	per query	A 60% B 40%	2,816.0 G	21.6
T4	Vision-language-action policy	per action-chunk step	A 60% B 40%	378.0 G	125.2
T5	Volumetric information-gain evaluation	per candidate viewpoint	C 100%	434,000	1.9
T5	Sampling-based graph + collision check	per replan	C 100%	16,000,000	1.6
T6	Learned SDF encoder (neural SDF-MPC)	per call	A 85% B 15%	22.0 G	406.4
T6	Learned navigation policy (DRL)	per call	A 85% B 15%	9.1 G	327.9
T6	Nonlinear MPC solve (quadrotor, 4 DoF)	per solve	C 100%	1,124,400	6.3
T6	Composite CBF safety filter	per solve	C 100%	29,130	3.2
T7	Attitude / rate or joint servo control	per update per DoF	C 100%	440	4.6

Ops per call is the forward-derived cost of one invocation at the configuration stated in the workbook. Mission demand is this figure multiplied by the update rate the profile assigns to that stage.

2 · COUNTING CONVENTIONS

Operations. One operation is one scalar arithmetic operation. A multiply-accumulate is two. This matters because published network costs are quoted inconsistently: the "GFLOPs" figure reported by Ultralytics and timm is a MAC count, and is doubled here. Transformer arithmetic is taken as two operations per parameter per token, which is the standard convention and is exact for the GEMM-dominated portion. Getting this wrong is a factor of two on the largest stages in the model, so it is stated explicitly in the workbook and applied uniformly.

Bytes. Bytes are counted at the DRAM interface, and cache-line granular wherever the access pattern is scattered: a four-byte read of a hashed voxel moves a sixty-four-byte line, and charging it four bytes would understate the memory system by an order of magnitude. Where a stage streams contiguously it is charged its actual bytes. All byte counts assume competent cache blocking and are therefore lower bounds — a poorly tiled implementation moves considerably more.

Precision classes. Each stage is split across three classes. Class A is INT8-eligible dense convolution and GEMM. Class B carries an FP16 floor: softmax, normalization, attention scores and regression heads. Class C carries an FP32/FP64 floor: factor graphs, signed-distance fields, model-predictive control, barrier-function filters. The split per stage and its justification are on the Precision classes sheet. A mixed stage is bound by its hardest component, not its dominant one.

Effective throughput. Class A is credited 2,000 GOP/s, Class B 300 GOP/s and Class C 15 GOP/s. These are the companion measurement annex’s stated assumptions, anchored to the fastest batch-1 result in the 360-configuration corpus and generous on B and C. They are inputs, not measurements of this workload, and they set every latency and occupancy figure in the model. Tripling the B and C figures is modeled in the sensitivity sheet.

3 · THE LATENCY AND OCCUPANCY MODEL

Arithmetic demand alone does not tell a systems engineer whether the loop closes. The model therefore converts every stage cost into a service time and an occupancy, using three definitions and no further assumptions.

Quantity	Definition, and what it is for
Service time	Seconds to execute one call of the stage with the entire machine to itself: the stage ops apportioned across the three precision classes and divided by each class effective throughput. It is a floor — it assumes no contention, no launch overhead and no cache cold-start.
Occupancy	Service time multiplied by the calls per second the profile requires. It is the fraction of one second of machine time the stage consumes. Occupancy at or above 1.00 means the stage cannot sustain its own update rate even alone on the machine, so it is a hard infeasibility rather than a degradation.
Oversubscription	The sum of occupancy across all stages in a profile: seconds of compute demanded per second of mission. A value of 16 means the machine must be sixteen times faster for the pipeline to run at its specified rates. This is the single number that expresses the compute deficit in the time domain.
Reactive chain latency	The sum of service times along the sense-to-act critical path — front-end, state estimation, distance field, learned policy, MPC, safety filter, servo. It is compared against the mission deadline, which is the glass-to-actuator budget the mission physics allows.

Because the deadline for a stage running at rate R is 1/R, occupancy and the per-stage constraint ratio are the same quantity under this model. That is deliberate: it removes a layer of allocated latency budgets that would otherwise be free parameters.

4 · WHAT THE MODEL RETURNS

Form factor	Mission capability	Sustained TOP/s	Sustained GB/s	Over-subscription	Reactive chain ms	Deadline ms	Stages that miss their own rate
Edge AI device	Event detection & classification	0.05	0	0.1	0	200	0
Edge AI device	Multi-stream tracking & anomaly	0.63	4	0.9	0	200	0
Edge AI device	Supervisory control & command grounding	0.80	27	1.3	0	500	0
Drone	Inspection (structure / asset)	1.76	24	3.2	131	120	1
Drone	ISR (endurance / wide-area search)	10.47	306	16.2	147	200	3
Drone	Interceptor (terminal engagement)	12.46	102	16.6	147	80	3
Quadruped	Inspection (plant walkdown)	3.71	32	5.2	111	100	1
Quadruped	Surveillance (persistent patrol)	6.53	38	8.1	111	100	1
Quadruped	ISR (dismounted, comms-denied)	10.48	198	15.3	148	150	3
AMR	Warehousing (structured aisles)	1.34	16	2.2	101	150	0
AMR	Logistics (mixed / dynamic yard)	5.09	35	6.9	111	120	1
AMR	Loading / unloading (manipulation)	11.32	93	16.2	111	100	3
Humanoid	Industrial automation (structured cell)	10.67	71	18.0	117	80	3
Humanoid	Cobot (human-adjacent, contact-rich)	18.24	131	28.9	117	60	5
Humanoid	House work (open-world, long-horizon)	25.06	347	42.4	154	100	7
Autonomous vehicle	SAE L2 / L2+ (partial automation)	0.95	14	2.7	0	150	1
Autonomous vehicle	SAE L3 (conditional automation)	6.44	99	13.4	181	100	5
Autonomous vehicle	SAE L4 / L5 (high / full automation)	25.06	386	48.0	181	100	8

Oversubscription is seconds of compute demanded per second of mission at the stated effective throughputs. Reactive chain is the sense-to-act critical path with the whole machine available; a dash means the profile has no closed control loop. Both are lower bounds.

5 · WHERE THE DEMAND COMES FROM

Two structural facts hold across all eighteen profiles and are worth stating because they determine which numbers a reviewer should care about. First, semantic perception — Tier 4 — is between 64% and 99% of the arithmetic everywhere, so the detector and language-model unit costs carry almost the entire result and the remaining seventeen are, arithmetically, rounding. Second, the reverse holds in the time domain: Class C work is under 3% of the operations in every profile and between 13% and 71% of the schedule, because it runs at fifteen effective GOP/s against two thousand.

The practical consequence for diligence is that the model has a small number of load-bearing inputs. If the detector op count, the detector rate, the language-model parameter count and the language-model query rate are accepted, the arithmetic totals follow within roughly ten percent regardless of what is assumed about the other fifteen stages. Those four inputs are the ones to interrogate, and they are also the four that are most directly checkable against published model specifications.

6 · VALIDATION

The model is checked against two independent references it was not fitted to. The first is the companion annex's published drone regimes, derived earlier by a separate stage-level analysis. The second is silicon actually fielded at each SAE level, which is the only place in embodied autonomy where a capability-to-compute mapping exists outside this work.

Check	Quantity	This model	Reference	Delta
Companion annex, far flight	Sustained TOP/s	10.47	10.24	+2.3%
Companion annex, air superiority	Sustained TOP/s	12.46	12.81	-2.7%
Companion annex, far flight	Oversubscription s/s	16.2	16.2	-0.1%
Companion annex, air superiority	Oversubscription s/s	16.6	17.3	-3.9%
SAE L2 / L2+	Nameplate band, TOPS	19–48	2.5–144	overlaps
SAE L3	Nameplate band, TOPS	129–322	24–254	overlaps
SAE L4 / L5	Nameplate band, TOPS	501–1,253	254–2,000	overlaps

Both drone regimes reproduce within three percent on arithmetic and within four percent on oversubscription, from a derivation that shares no coefficient with the earlier one. The three SAE bands each overlap the range of silicon fielded at that level. Neither check was used to set any unit cost, and both are reproduced on the Validation sheet of the workbook.

7 · EXPOSURE REGISTER

Seven of the nineteen stages disagree with the value implied by the earlier published model by more than a factor of two. Two are reparameterizations that leave sustained demand unchanged. The other five are all cases where this derivation is the cheaper of the two, and all are under half a percent of any mission total — correcting them upward would raise the demand figures, not lower them.

Stage	Disagreement	Max share of a total	Assessment
Stereo SGM	8.0x higher per frame	4.5%	A reparameterization, not a disagreement. This model itemizes a 128-disparity frame and runs it at 40 Hz; the earlier model implied a cheaper frame at roughly 300 Hz. Sustained demand agrees within 7% and the frame rate here is the defensible one.
Radar FMCW	2.3x higher per scan	0.9%	Aperture. This model states a 32-channel 1024x256 MIMO radar at 20 Hz. Sustained demand differs by 6%. Lighter profiles use a 12-channel radar, costed separately.
Gain evaluation	177x lower	0.01%	A candidate viewpoint at 200 rays by 120 voxels is 0.43 MOP. We believe the earlier figure was too high; the difference does not move any conclusion.
Graph + collision	16x lower	0.02%	Two thousand samples with fifty collision queries per edge is 16 MOP.
Nonlinear MPC	22x lower	0.47%	A thirty-step Riccati recursion at twelve states is 1.1 MOP. It matters only for the humanoid, where sixty states raise it by two orders of magnitude and it is modeled explicitly rather than scaled.
CBF safety filter	172x lower	0.00%	A three-variable QP over twelve constraints is genuinely small; the cost is the distance-field sweep. Its significance is its FP64 floor and kilohertz rate, not its op count.
Servo control	2,270x lower	0.00%	A cascaded PID is a few hundred operations and runs on the autopilot or joint microcontroller, not the mission computer.

8 · WHAT IS NOT MODELED

Five categories of cost are excluded. All five add demand; none subtracts, so the totals in this model are lower bounds on what a fielded system requires.

Kernel launch and scheduling overhead. Excluded entirely. The MPC and barrier-function stages are serial chains of small dependent kernels, and at a few microseconds of dispatch each, a long SQP recursion can spend more time being dispatched than computing. This is the largest single omission and it runs against the incumbent architecture.

Inter-process communication. Real stacks pass messages between nodes with serialization, copies and queueing. None of that is counted here.

Cold caches and contention. Service times assume each stage owns the whole machine with a warm cache. In a running system the stages contend, so measured latency exceeds these figures.

Sensor drivers, thermal throttling and allocation. Driver-level cost, sustained-load clock reduction and memory management are all outside the model.

Redundancy beyond what is stated. Only the SAE L4/L5 profile carries an explicit dual-redundant stack. Safety-certified variants of the other profiles would carry more.

9 · SENSITIVITY: THE FOUR INPUTS THAT MATTER

Input	Effect of doubling	Why it dominates
Detector rate and op count	+64% to +99% of total	Tier 4 is 64–99% of the arithmetic in every profile. This is the single largest lever and the most directly checkable against published model specifications.
Language-model parameters or query rate	up to +63% of total	Dominant wherever an onboard VLM or VLA is carried; it also drives the bandwidth column, because decode re-reads the whole weight set per token.
Achieved-to-peak ratio (2–5%)	halves nameplate silicon	Does not change sustained demand at all, only the silicon required to deliver it. The SAE validation is the main external evidence that the band is right.
Class B and C effective throughput	oversubscription 17x → ~6x	Changes no arithmetic figure, only latency and occupancy. Tripling both does not remove the deficit from any profile that has one.

10 · REPRODUCING THIS MODEL

The model is four Python files totaling under a thousand lines, and the workbook is generated from them rather than maintained by hand. `derive.py` holds the itemized unit-cost derivations; `pipeline.py` holds the cost, class, service-time and occupancy model; `profiles.py` holds the eighteen mission specifications as explicit sensor suites and update rates; `book.py` emits the workbook. Changing a single rate in `profiles.py` and re-running regenerates every figure in the document series, which is the property that makes a reviewer’s disagreement cheap to test rather than a matter of argument.

Companion documents. “The Compute Requirements of Embodied Autonomy” states the argument this annex supplies the inputs for. “Measured Energy Efficiency of Embedded AI Compute” documents the 360-configuration measurement campaign that supplies the effective-throughput and achieved-to-peak assumptions. “Compute Requirements by Robot Form Factor and Mission Capability” presents the eighteen-profile result. The accompanying workbook, `BranesAI-Workload-Data-Annex.xlsx`, is the model itself. branes.ai